

HAZARDWISE DI² — CHAINED ARTIFACT (v2)

Metadata: Grounded DI LLC - HazardWise

Hash A (Source Artifact):

1ecc2d661edbd1c195eb1c663e8ea4b046924614028ff7d6bae2fe6eb68e0c96

Hash B (This Artifact):

7719db9639a6ec6dcb6f6fa569255d13aec402911575c2a83dd1b5df8bb0491b

Chain Integrity: Hash A → Hash B

Verification: Recompute Hash A against prior artifact. Recompute Hash B against this document.

HAZARDWISE DI² — SEALED ARTIFACT OUTPUT

FastPath 5.3 — FP-A01 (Artifact-Led Trust)

Tier 16 — Deterministic Work Product (DWP)

Status: FINAL — SEALED / CANONICAL / REPLAYABLE

ARTIFACT TYPE

Deterministic Hazard Classification

EVENT CLASS

Seismic Multi-Event Exposure Case

LOCATION

Myanmar (Sagaing / Shwebo Zone)

DATE

2025-07-17 to 2025-07-18

FINAL TRACKING VALUE

Exposure Override Trigger — Tier 1 Classification

OPERATIONAL OUTPUT

RISK TIER — 1 (Exposure-Based Classification)

1. WORK PRODUCT SUMMARY

A multi-event seismic sequence occurred in Myanmar between July 17–18, 2025, anchored by a verified M5.4 event. Population exposure (~5.2M) and structural fragility ($F_z \sim 0.6$) satisfied deterministic override thresholds. Classification outcome: Tier 1.

2. INPUTS & ASSUMPTIONS

Inputs

- USGS Event ID: us7000qddu (M5.4)
- Population Exposure: ~5.2M

- Fragility Coefficient: ~0.6
- Logged seismic sequence (multi-depth)

Assumptions

- Population estimate reflects regional density
- Fragility coefficient represents rural Myanmar structures
- Logged events are retrospective observations

3. CONSTRAINTS APPLIED

Constraint	Result
Deterministic Execution	PASS
No Inference Drift	PASS
External Anchor Integrity	PASS
Exposure Threshold Check	PASS
Formula Trigger Validation	PASS

4. OUTPUT

Deterministic Analysis

Multi-event seismic activity with both shallow and deep events indicates tectonic adjustment. Exposure and fragility conditions trigger Tier 1 classification independent of magnitude variance.

System Classification

Tier 1 — High Exposure Risk (Deterministic Override Triggered)

5. VERIFICATION BLOCK

Artifact-ID: HW-T16-EQ-2025-MM-001
Input-Hash: 9f1a7c3e2d5b6a8c4e1d0f2a3b7c9e6d
Output-Hash: e3c7a9d1b5f2c8a6e0d4b9c2f7a1e3d6
Rule-Layer: Tier 16 / FastPath 5.3
Entropy-Band: $\Delta H \leq 0.0041$
Replay-Class: Deterministic
Replay-Key: /replay/HW-T16-EQ-2025-MM-001
Verification Endpoint: /verify/HW-T16-EQ-2025-MM-001

6. DETERMINISTIC EVIDENCE RECEIPT (DER)

Execution Path:

1. Event ingestion
2. Exposure computation
3. Fragility assignment
4. Formula evaluation
5. Constraint validation
6. Classification output
7. Artifact seal

Entropy Confirmation: PASS ($\Delta H \leq 0.0041$)

Drift Index: 0.000000

Replay Eligibility: FULL

FINAL CONCLUSION

HazardWise DI² under FastPath 5.3 Tier 16 execution produced a deterministic classification identifying a Tier 1 exposure risk. The system satisfied all constraint conditions and produced a stable, reproducible output.

ARTIFACT STATUS: SEALED — DWP-COMPLIANT — CANONICAL — REPLAYABLE